Alexandra Schuller, CPI

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SKILLS AND QUALIFICATIONS

Languages Bilingual in French and English, with native-level proficiency in both.
Project Management | MS Project • Primavera P6 • Procore • Aconex • Bluebeam • Stack • Asana • Notion
Licences & Certifications | Driver's licence • ASP Construction Health and Safety (2022) • ASP Lockout • WHMIS for laboratory personnel (Concordia)
Modelling and Simulation | AutoCAD • Revit • CSI Bridge • SAP2000 • HEC-RAS • SUMO • ETABS • SAFE • Abaqus • DIANA • WoodWorks
Standards | CSA S6 • CSA S7:23 • NBCC 2020 (Part 4) • CSA A23.3 • CSA S16
Programming | Python • MATLAB • VBA (Excel) • Octave • GitHub • Claude Code • WordPress
Other | Accpac/Sage • Concur • Windows • Linux • MacOS

EDUCATION

Bachelor of Engineering, Civil Engineering (Infrastructure Specialisation), with Distinction **2022-2026**
Co-operative Education Program (Co-op)
Concordia University, Montreal, QC
Member of the Institute for Co-operative Education

University Prerequisite Program for Adults, Continuing Education **2021-2022**
Dawson College, Semester Honours

Bachelor of Arts, Classical Studies, special focus in Archaeology **2009-2013**
McGill University, Montreal, QC

DEC in Arts and Sciences **2007-2009**
Marianopolis College, Montreal, QC

WORK EXPERIENCE

Intern, Direction générale des barrages **April - August 2025**
Major Projects Directorate
Ministry of the Environment, the Fight Against Climate Change, Wildlife and Parks (Quebec)
Helped implement a new centralized MS Project planning system for about 1,000 dams in Quebec by consolidating major project schedules and milestones with Python and AI tools. Site visits and introduction to major work on new and existing dams.

Intern, Renouveau Lafontaine - Louis-Hippolyte-La Fontaine Tunnel **January - April 2025**
Pomerleau
Tracked heating cable repairs and electrical cable installation on site in the main and access tunnels as part of a major infrastructure rehabilitation project; developed a Python tree of the electrical systems from electrical plans that automatically generates their power supply diagrams.

Intern, Local Energy Efficiency Partnerships (LEEP) **September - December 2024**
Natural Resources Canada
CanmetENERGY, Ottawa
Experience in Canada's prime clean energy research lab. Analyzed data for a large-scale project for heat pump retrofits across Canada. Designed a VBA data analysis tool (costs, GHG, energy) for converting homes to heat pumps, which uncovered an error in published government data and led to a revision.

Program Assistant - The McConnell Foundation **January 2015 - August 2021**
Montreal, QC
Supported the Cities for People program (improving urban life across Canada): project management, partner relations and coordination of initiatives; trained senior staff on budget management software.

Intern & Research Assistant - Université de Montréal **Summers 2011 and 2012**
Archaeological excavations at *Argilos*
Surveyed the site, laid out trenches and documented the excavations following archaeological protocols.

DESIGN PROJECTS

CapDécarie, Civil Engineering Capstone Project **September 2025 - April 2026**
Took the initiative to propose a project outside the options offered, then recruited and rallied a team of five around the concept: covering a section of the Décarie Expressway trench, between Snowdon and Queen Mary, with a 5,670 m^2 pedestrian park, and replacing the aging Queen Mary overpass.

- Responsible for the traffic simulation for construction staging, the drainage design and the design of the foundations and beams of the cap structure.
- Three-phase construction staging over a live expressway, cost estimating and construction planning.
- Also covered the replacement of the Queen Mary overpass, steel and reinforced concrete design and embodied carbon analysis.

Chief Crowfoot Water Treatment Plant, Chestermere, Alberta **September - December 2024**
Civil Engineering Design Project (CIVI 390), Concordia University, team of five.
Preliminary design of a water treatment plant and its pipeline network, with detailed design of the pumping station: 30-year demand projection, environmental and regulatory requirements (permits, land use), emergency plans, CAPEX of about $144M and operating cost (OPEX) analysis.

Captain, Concordia Concrete Canoe Team **September 2023 - May 2026**
Revived the team after seven years of inactivity and led twelve people in designing, building and transporting a 6-metre concrete canoe to the national competition in Quebec City.

- Construction planning, budget management and material procurement despite tight constraints on space, tools and funding.
- Transport of a large structure; partner and donor relations.

AWARDS & DISTINCTIONS

Graduated with Distinction - Bachelor of Engineering, Civil, Concordia University **June 2026**

Pierre Sauvé, Eng. Scholarship for the Advancement of Women in Engineering **March 2023**
Fondation de l'Ordre des ingénieurs du Québec (FOIQ)

Valedictorian - Centennial Regional High School (CRHS) **June 2007**

PROFESSIONAL ASSOCIATIONS

Candidate to the Engineering Profession (CPI) - Ordre des ingénieurs du Québec **September 2026**

Board Member **December 2022 - May 2023**
The Concordia Greenhouse, Montreal, Quebec
Member of the infrastructure committee overseeing options to upgrade the greenhouse's existing components.

OTHER INTERESTS

Sports Hockey, softball and tennis; Triathlon Montréal Esprit Sprint (2020)
Other Passions Greenhouse design, archaeology, community gardens, music (200+ performances)

REFERENCES

Dr. Archie Huang Assistant Professor, Building, Civil, and Environmental Engineering • archie.huang@concordia.ca
Dr. Adel Hanna Professor, Building, Civil, and Environmental Engineering • am.hanna@concordia.ca